

where we used that $F(s, t) \leq 1$. For the second term of (30a) we have due to the coordinate change from post- to pre-collisional states

$$\begin{aligned} & \varepsilon \mu_{1,2} \lambda_{1,2} \int_{\mathbb{R}} \left| \int_0^t F(r, t) \int_{\mathbb{R}^2} \int_0^\infty \left(S_3(s) (f_1 f_2 - \tilde{f}_1 \tilde{f}_2) \right) (v_1, v_2, v_3) ds dv_3 dv_2 dr \right| dv_1 \\ & \leq \varepsilon \lambda_{1,2} \int_0^t \int_{\mathbb{R}^3} F(r, t) |f_1 f_2 - \tilde{f}_1 \tilde{f}_2| dv_1 dv_2 dv_3 dr \\ & \leq \varepsilon \lambda_{1,2} \int_0^t \int_{\mathbb{R}^3} \left(|f_1 - \tilde{f}_1| |f_2 - \tilde{f}_2| |\tilde{f}_1| \right) dv_1 dv_2 dv_3 dr \\ & \leq \varepsilon t \lambda_{1,2} \sup_{0 < r < t} \left(M_2(r) \|f_1(\cdot, r) - \tilde{f}_1(\cdot, r)\|_{L^1(\mathbb{R})} + M_1(r) \|f_2(\cdot, \cdot, r) - \tilde{f}_2(\cdot, \cdot, r)\|_{L^1(\mathbb{R}^2)} \right), \end{aligned}$$

where we further used the trivial identity $f_1 f_2 - \tilde{f}_1 \tilde{f}_2 = (f_1 - \tilde{f}_1) f_2 + (f_2 - \tilde{f}_2) \tilde{f}_1$. Similar computations can be done for the right-hand-side of (30b). Indeed, we have

$$\begin{aligned} & \frac{\lambda_{1,1}}{\varepsilon} \int_{\mathbb{R}^2} \left| \int_0^t \left(S_2((t-s)/\varepsilon) (f_1 f_1 - \tilde{f}_1 \tilde{f}_1) \right) ds \right| dv_1 dv_2 \\ & \leq \frac{\lambda_{1,1}}{\varepsilon} \int_{\mathbb{R}^2} \left| \int_0^t e^{-\mu_{1,1} \frac{(t-s)}{\varepsilon}} - \lambda_{1,2} \int_0^t M_1(q) dq (f_1 f_1 - \tilde{f}_1 \tilde{f}_1) ds \right| dv_1 dv_2 \\ & \leq 2t \frac{\lambda_{1,1}}{\varepsilon} \sup_{0 < s < t} M_1(s) \sup_{0 < s < t} \|f_1(\cdot, s) - \tilde{f}_1(\cdot, s)\|_{L^1(\mathbb{R})}, \end{aligned}$$

and

$$\begin{aligned} & \mu_{2,1} \lambda_{1,2} \int_{\mathbb{R}^2} \left| \int_0^t \left(S_2((t-s)/\varepsilon) \int_{\mathbb{R}} \int_0^\infty \left(S_3(r) (f_1 f_2 - \tilde{f}_1 \tilde{f}_2) \right) (\cdot, \cdot, v_3) dr dv_3 \right) (v_1, v_2) ds \right| dv_2 dv_1 \\ & = \lambda_{1,2} \int_{\mathbb{R}^2} \left| \int_0^t \int_{\mathbb{R}} e^{-\mu_{1,1} \frac{(t-s)}{\varepsilon}} - \lambda_{1,2} \int_0^t M_1(q) dq (f_1 f_2 - \tilde{f}_1 \tilde{f}_2) dv_3 ds \right| dv_2 dv_1 \\ & \leq t \lambda_{1,2} \sup_{0 < s < t} \left(M_1(s) \|f_2(\cdot, \cdot, s) - \tilde{f}_2(\cdot, \cdot, s)\|_{L^1(\mathbb{R}^2)} + M_2(s) \|f_1(\cdot, s) - \tilde{f}_1(\cdot, s)\|_{L^1(\mathbb{R}^2)} \right). \end{aligned}$$

These estimates can be made contractive if t is chosen small enough, giving local existence. Convergence of $(M_1(t), M_2(t))$ to (M_1^∞, M_2^∞) as $t \rightarrow \infty$ (see (21)) allows to iterate the contraction estimate and implies a global L^1 -bound of (f_1, f_2) and therefore global existence. $\square$

Remark 3.1. We notice that the third estimated term, part of the right-hand-side of f_2 , shows a term with constant proportional to $\frac{t}{\varepsilon}$. This is a consequence of the fact that the dynamics of f_2 are much faster than the ones of f_1 and indicates that time-uniform estimates for the fast variable f_2 break down for vanishing ε . Clearly the limits $\varepsilon \rightarrow 0$ and $t \rightarrow 0$ do not commute. Thus, we expect difficulties arising taking the the limit $\varepsilon \rightarrow 0$ for small time t .

3.3. Instantaneous limit. We consider the mild formulation (30b) for f_2^ε which after the coordinate change $s \mapsto t - \varepsilon s$ reveals its time-delay structure

$$\begin{aligned} f_2^\varepsilon(v_1, v_2, t) &= (S_2(t/\varepsilon) f_2^I)(v_1, v_2) + \lambda_{1,1} \int_0^{t/\varepsilon} \left(S_2(s) f_1^\varepsilon(\cdot, t - \varepsilon s) f_1^\varepsilon(\cdot, t - \varepsilon s) \right) (v_1, v_2) ds \\ &\quad + \varepsilon \mu_{2,1} \lambda_{1,2} \int_0^{t/\varepsilon} \left(S_2(s) \int_{\mathbb{R}} \int_0^\infty \left(S_3(r) f_1^\varepsilon(\cdot, t - \varepsilon s) f_2^\varepsilon(\cdot, \cdot, t - \varepsilon s) \right) (v_1, v_2, v_3) dr dv_3 \right) ds. \end{aligned}$$

Taking the formal limit $\varepsilon \rightarrow 0$ we obtain

$$(32) \quad f_2(v_1, v_2, t) = \lambda_{1,1} \int_0^\infty \left(S_2(s) f_1(\cdot, t) f_1(\cdot, t) \right) (v_1, v_2) ds.$$

Now substituting it in the equation for f_1 yields at leading order

$$(33) \quad \partial_t f_1 = 2\lambda_{1,1} \mu_{1,1} \int_{\mathbb{R}} \int_0^\infty \left(S_2(s) f_1(\cdot, t) f_1(\cdot, t) \right) ds dv_2 - 2\lambda_{1,1} f_1 \int_{\mathbb{R}} f_1 dv_2,$$